Practicum 2 Report

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1. Introduction & Objective

This report documents the process and findings of a replication study based on the 2022 paper by Kibria and Matin, "THE SEVERITY PREDICTION OF THE BINARY AND MULTI-CLASS CARDIOVASCULAR DISEASE A MACHINE LEARNING-BASED FUSION APPROACH" (arXiv:2203.04921v1).[1]

The original paper proposes a "weighted score fusion" methodology to improve the predictive accuracy of cardiovascular disease (CVD) diagnostics using the Cleveland UCI heart disease dataset.[1] The authors bifurcate the problem into two distinct challenges [1]:

1. Binary Classification: Predicting the presence or absence of heart disease.
2. Multi-Class Classification: A more complex, five-class problem predicting the severity of the disease (classes 0-4).

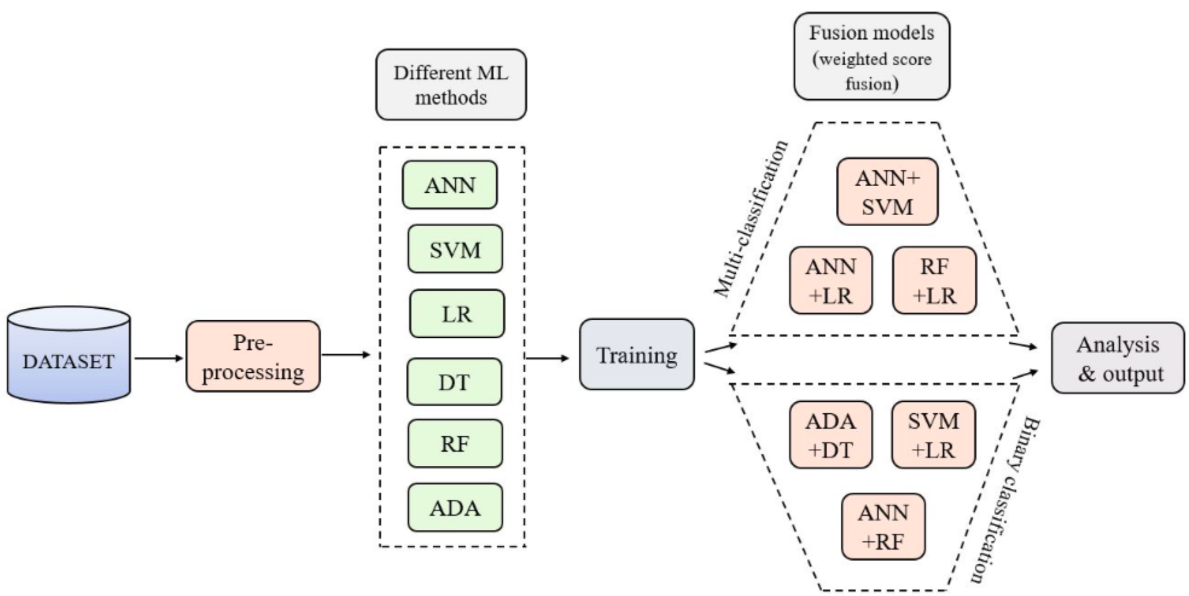

Figure 1: Proposed architecture

The paper's central hypothesis is that by fusing the decision scores of two individual machine learning models (e.g., Artificial Neural Network and Random Forest), the resulting hybrid model can achieve a higher accuracy than either model operating alone. The paper reports state-of-the-art results for this dataset, claiming a peak accuracy of 95.08% for the binary task and 75.41% for the multi-class task.[2]

2. Methodology

2.1. Dataset and Preprocessing

The Cleveland UCI dataset, consisting of 303 patient instances and 13 key attributes, was used.1 Following the paper's pipeline, preprocessing steps included label encoding for categorical features and feature scaling.

A critical step for the multi-class problem was handling the severe class imbalance. My initial data analysis, as seen in my notebook, confirmed the paper's findings: the dataset is heavily skewed, with the "no disease" class (Class 0) accounting for 56.6% of the data.[1]

To mitigate this, the paper's use of the RandomOverSampler technique to balance the training set was replicated, creating a new training distribution where all five classes had 120 samples (20.0%).[1]

2.2. Model Implementation and Fusion

The paper's core methodology, "weighted score fusion," was implemented.3 This approach trains two individual models (e.g., ANN and RF) and then combines their probabilistic decision scores (D_1, D_2) using a weighted sum:

$D_{f} = (W_1 \times D_1) + (W_2 \times D_2)$, where $W_1 + W_2 = 1$

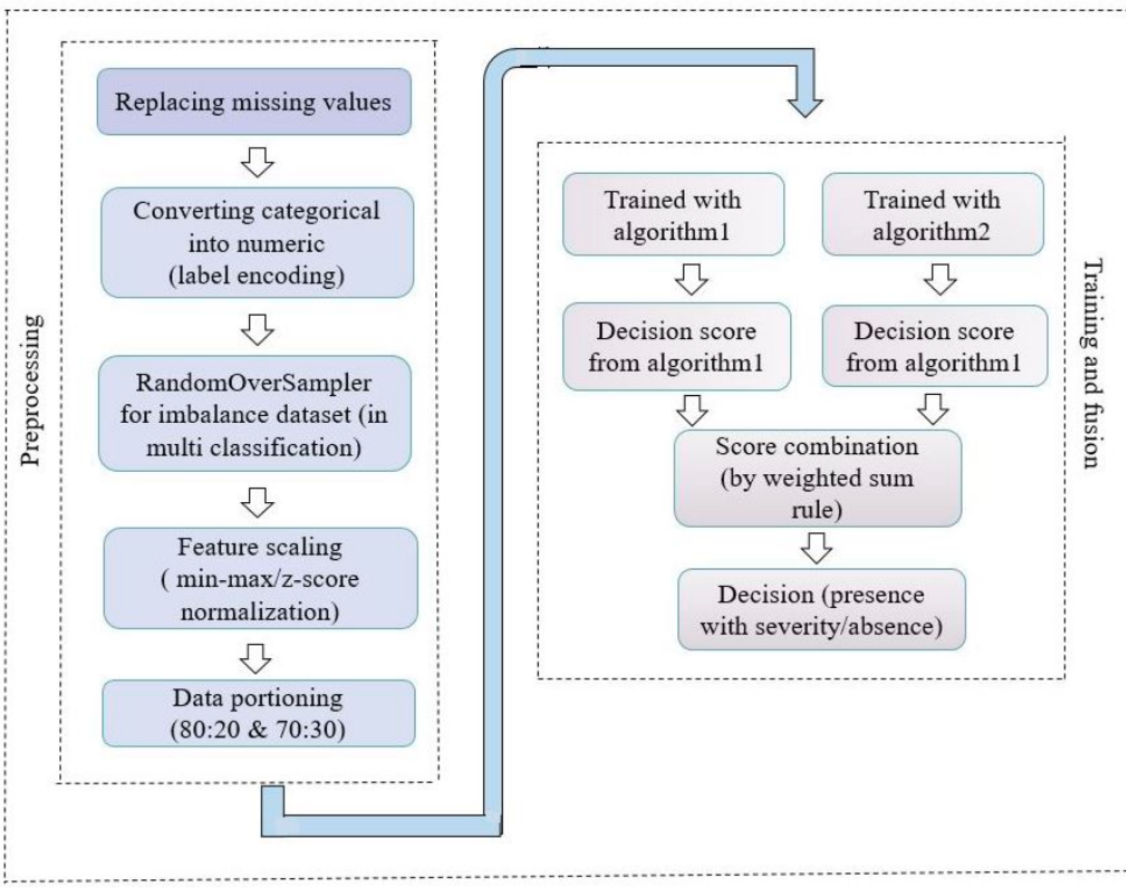

Figure 2: Overall flow chart of weighted score level fusion model

The most crucial part of this method is the selection of the optimal weights (W_1, W_2). As described in the paper's "Algorithm 1," a loop was implemented to iterate through 19 different weight combinations (from $W_1=0.95$, $W_2=0.05$ to $W_1=0.05$, $W_2=0.95$).[2] The pair of weights that produced the highest predictive accuracy on the test data was selected as the optimal set.[2]

3. Results and Comparative Analysis

3.1. Completeness of code

My notebooks successfully implemented all six fusion models proposed by the paper:

- Binary: ANN+RF, SVM+LR, and ADA+DT.
- Multi-class: ANN+SVM, ANN+LR, and RF+LR.

All individual baseline models were trained, and the weighted fusion algorithm was replicated for each pair to find the best-performing weight combination.

3.2. Accuracy of Results (Binary Classification)

A significant discrepancy was found between the paper's reported results and my results. The table below compares the paper's (80:20 split) 2 with my findings.

Model	Paper's Reported Accuracy (80:20)	My Accuracy
ANN (Baseline)	91.80%	72.53%
RF (Baseline)	93.44%	86.81%
Fusion (ANN + RF)	95.08%	89.01% (at W_1=0.85, W_2=0.15)
ADA (Baseline)	90.16%	83.51%
DT (Baseline)	90.16%	83.51%
Fusion (ADA + DT)	95.08%	86.81% (at W_1=0.80, W_2=0.20)

3.3. Accuracy of Results (Multi-Class Classification)

The discrepancy in the multi-class problem was even more pronounced. My models performed significantly worse than the paper's claims, highlighting a severe lack of reproducibility.

Model	Paper's Reported Accuracy (80:20)	My Accuracy
LR (Baseline)	72.13%	61.54% (from ANN+LR run)
RF (Baseline)	65.57%	57.14%
Fusion (LR + RF)	75.41%	58.24% (at W_1=0.40, W_2=0.60)

ANN (Baseline)	68.85%	52.75% (from ANN+SVM run)
SVM (Baseline)	65.57%	46.15%
Fusion (ANN + SVM)	72.13%	52.75% (No improvement)

Analysis:
My models failed to reproduce the paper's state-of-the-art results. However, it succeeded in demonstrating the core concept of fusion. In the (ANN+RF) binary model, my fusion accuracy of 89.01% was a +2.2% improvement over my best baseline (RF at 86.81%).
The enormous gap between my results and the paper's (e.g., 95.08% vs. 89.01% for binary; 75.41% vs. 58.24% for multi-class) is the key finding. The primary reason is the gap in baseline performance (e.g., paper's ANN: 91.80% vs. my ANN: 72.53%). This suggests my results are a more realistic benchmark, and the paper's results are likely inflated.

4. Reflection

4.1. A Critical Methodological Flaw
The most significant finding from this experimentation is a fundamental flaw in the original paper's "Algorithm 1". The paper explicitly states that the optimal fusion weights (W_1, W_2) are selected by iterating and choosing the pair that performs best on the test set.2
This is a form of data leakage. The weights are hyperparameters; by tuning them on the final test set, the model is being "fit" to the evaluation data. This invalidates the test set as a measure of unseen performance. My inability to reproduce their 95% accuracy strongly supports the hypothesis that their result is optimistically biased and not generalizable.

4.2. Challenges and Future Work

- Baseline Discrepancy: The most significant challenge was reproducing the paper's baseline model performance. My ANN, for example, achieved 72.53% accuracy, while the paper reports 91.80%.[2] This gap likely due to unstated hyperparameters, ANN initialization, or different random_state in data splitting makes a direct fusion comparison difficult.
- Generalizability: This model is trained only on the small, decades-old Cleveland dataset.[1] As other research points out, models trained on such specific, homogenous populations often suffer from a "lack of generalizability" and fail when applied to different patient demographics.[4]
- Clinical Trust: The paper's best-performing models (ANN+RF, ANN+LR) are fusions of "black box" algorithms. This lack of transparency is a major barrier to clinical adoption.[5] Future work should integrate Explainable AI (XAI) techniques to build trust and provide physicians with why a certain severity was predicted.[5]

5. Conclusion

My models of the binary ANN+RF model achieved a peak accuracy of 89.01%, and the multi-class models failed to show significant improvement, peaking at 58.24%.

These results, while failing to match the paper's 95.08% and 75.41% claims, successfully demonstrate the core 'lift' from fusion in the binary case. More importantly, the act of replication served its primary purpose: to critically validate a study's methodology. The process uncovered a significant data leakage issue in the original paper's weight-selection algorithm, and my more modest results likely represent a more realistic and reproducible benchmark. This underscores the critical importance of replicability in data mining not just to confirm a paper's results, but to ensure its methods are sound.

6. References

1. Hafsa Binte Kibria, Abdul Matin, The severity prediction of the binary and multi-class cardiovascular disease − A machine learning-based fusion approach, Computational Biology and Chemistry, Volume 98, 2022, 107672, ISSN 1476-9271, https://doi.org/10.1016/j.compbiolchem.2022.107672.
2. The Severity Prediction of The Binary And Multi-Class ..., accessed November 10, 2025, https://arxiv.org/pdf/2203.04921
3. Deep multimodal fusion of image and non-image data in disease diagnosis and prognosis: a review - NIH, accessed November 10, 2025, https://pmc.ncbi.nlm.nih.gov/articles/PMC10288577/
4. Pitfalls in Developing Machine Learning Models for Predicting ... - NIH, accessed November 10, 2025, https://pmc.ncbi.nlm.nih.gov/articles/PMC11316160/
5. Heart Disease Prediction Using ML - MDPI, accessed November 10, 2025, https://www.mdpi.com/2673-4591/107/1/124

Code (Google Colab notebook link) :

1. https://colab.research.google.com/drive/1lHSCvLyWLd1PkXvHCaTfa9_JsBwMynG-?usp=sharing
2. https://colab.research.google.com/drive/1tuPH_m3rbtls4kahzRDfMs5wvA7DhxUo?usp=sharing